Top Python Coding Questions

@Tajamulkhann

Beginner

EVEN OR ODD CHECKER

```python
# Check if a number is even or odd
num = int(input("Enter a number: "))
print("Even" if num % 2 == 0 else "Odd")
```

REVERSE A STRING

```python
# Reverse a string using slicing
s = "Python"
print(s[::-1])  # Output: "nohtyP"
```

FIBONACCI SERIES GENERATOR

```python
# Generate Fibonacci series up to n terms
n = int(input("Enter the number of terms: "))
a, b = 0, 1
for _ in range(n):
    print(a, end=" ")
    a, b = b, a + b
```

CHECK LEAP YEAR

```python
# Check if a given year is a leap year
year = int(input("Enter a year: "))
if (year % 4==0 and year % 100!=0) or (year % 400==0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

PALINDROME CHECKER

```python
# Function to check if a string is a palindrome
def is_palindrome(s):
    return s.lower() == s.lower()[::-1]
print(is_palindrome("madam")) # Output: True
```

Basic

LIST COMPREHENSION CHALLENGE

```python
# Generate squares using list comprehension
squares = [x**2 for x in range(1, 11)]
print(squares) # Output: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

ARMSTRONG NUMBER CHECKER

```python
def is_armstrong(n):
    num_str = str(n)
    power = len(num_str)
    return n == sum(int(digit) ** power for digit in num_str)
print(is_armstrong(153)) # Output: True
```

LAMBDA FUNCTION USAGE

```python
# Lambda function to compute square of a number
square = lambda x: x ** 2
print(square(5)) # Output: 25
```

FIZZBUZZ IMPLEMENTATION

```python
# Classic FizzBuzz problem
for i in range(1, 101):
    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")
    elif i % 3 == 0:
        print("Fizz")
    elif i % 5 == 0:
        print("Buzz")
    else:
        print(i)
```

MAP FUNCTION IMPLEMENTATION

```python
# Using map() to double values in a list
nums = [1, 2, 3, 4]
doubled = list(map(lambda x: x * 2, nums))
print(doubled) # Output: [2, 4, 6, 8]
```

Intermediate

FILTER FUNCTION USAGE

```
# Filter even numbers using filter()
nums = [1, 2, 3, 4, 5, 6]
evens = list(filter(lambda x: x % 2 == 0, nums))
print(evens) # Output: [2, 4, 6]
```

REMOVE DUPLICATES FROM LIST

```
# Remove duplicates from a list
duplist = [1, 2, 3, 4, 4, 5, 5]
unique_list = list(set(duplist))
print(unique_list) # Output: [1, 2, 3, 4, 5]
```

VOWEL COUNTER

```
# Function to count vowels in a given string
def count_vowels(s):
    return sum(1 for char in s.lower() if char in "aeiou")
print(count_vowels("Hello World")) # Output: 3
```

ANAGRAM CHECKER

```
def check_anagram(s1, s2):
    return sorted(s1.lower()) == sorted(s2.lower())
print(check_anagram("listen", "silent")) # Output: True
```

MISSING NUMBER FINDER

```
def find_missing(nums, n):
    return n * (n + 1) // 2 - sum(nums)
print(find_missing([1, 2, 4, 5], 5)) # Output: 3
```

Advanced

CHARACTER FREQUENCY IN A STRING

```
def char_frequency(text):
    freq = {}
    for char in text.lower():
        freq[char] = freq.get(char, 0) + 1
    return freq
print(char_frequency("Hello World"))
# Output: {'h':1, 'e':1, 'l':3, 'o':2, ' ':1, 'w':1, 'r':1, 'd':1}
```

REVERSE WORDS IN A SENTENCE

```
def reverse_words(sentence):
    words = sentence.split()
    reversed_words = [word[::-1] for word in words]
    return " ".join(reversed_words)
print(reverse_words("Hello World"))  # Output: "olleH dlroW"
```

COMMON ELEMENTS OF TWO LISTS

```
list1 = [1, 2, 3, 4]
list2 = [3, 4, 5, 6]
print(list(set(list1) & set(list2))) # Output: [3, 4]
```

ITERATE OVER A DICTIONARY

```
fruits = {"apple": 2, "banana": 3, "cherry": 1}
total_quantity = 0
for fruit, quantity in fruits.items():
    print(fruit, quantity)
    total_quantity += quantity
print("Total:", total_quantity) # Output: apple 2, banana 3, cherry 1, Total : 6
```

REMOVE DUPLICATES

```
my_list = [1, 2, 3, 4, 2, 5, 3, 6, 3]
from collections import Counter
count = Counter(my_list)
duplicates = {item: freq for item, freq in count.items() if freq > 1} # {2: 2, 3: 3}
```

Expert

PRINT PYRAMID PATTERN

```
n = 5
for i in range(1, n + 1):
    print(" " * (n - i) + "*" * (2 * i - 1))
```

SECOND LARGEST ELEMENT

```
def second_largest(nums):
    nums = list(set(nums))  # Remove duplicates first
    nums.sort(reverse=True)
    return nums[1] if len(nums) > 1 else None
print(second_largest([10, 20, 40, 30]))  # Output: 30
```

NUMBER REVERSAL

```
def reverse_number(n):
    reversed_num = 0                # Start with 0 as the reversed number
    while n > 0:                    # Repeat until no digits left in n
        digit = n % 10              # Get the last digit of n
        reversed_num = reversed_num * 10 + digit  # Add digit to the reversed number
        n = n // 10                 # Remove the last digit from n
    return reversed_num             # Return the reversed number
print(reverse_number(12345))       # Output will be 54321
```

TWO SUM PROBLEM

```
def two_sum(nums, target):
    seen = {}                               # Dictionary to store number and its index
    for i, num in enumerate(nums):          # Loop through the list with index
        complement = target - num           # Find the number needed to reach the target
        if complement in seen:              # If we've already seen the complement
            return [seen[complement], i]    # indices of complement and current number
        seen[num] = i                       # Store the current number with its index
    return []                               # Return empty list if no pair found
# Example usage
print(two_sum([2, 7, 11, 15], 9))          # Output: [0, 1]
```

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